WORKED SOLUTION

STEP I 1989, Question 8

Pure • Multiple Angle Formulae

> **STEP I 1989, Question 8**
>
> Read the question before you go on. It is in the STEP Support Programme's question database:
>
> step.maths.org/questions/89-s1-q8

The key idea

De Moivre's theorem turns $\cos n\theta$ into the real part of $(\cos\theta + \mathrm{i}\sin\theta)^n$, and replacing $\sin^2\theta$ by $1 - \cos^2\theta$ leaves a polynomial in $\cos\theta$. For the final part, the sextic looks like a combination of the two polynomials just found, so we look for one: it turns out to be half of $\cos 6\theta - \cos 4\theta$, and solving $\cos 6\theta = \cos 4\theta$ is easy.

Solution

Throughout, write $c = \cos\theta$ and $s = \sin\theta$, so $s^2 = 1 - c^2$.

Part (i)

By de Moivre's theorem, $\cos 4\theta + \mathrm{i}\sin 4\theta = (c + \mathrm{i}s)^4$. Expanding by the binomial theorem and keeping the real terms (those with even powers of $\mathrm{i}s$),

$$\begin{aligned}\cos 4\theta &= c^4 - \tbinom{4}{2}c^2s^2 + s^4 \\ &= c^4 - 6c^2(1 - c^2) + (1 - c^2)^2 \\ &= c^4 - 6c^2 + 6c^4 + 1 - 2c^2 + c^4 \\ &= 8c^4 - 8c^2 + 1,\end{aligned}$$

which is the required identity.

Part (ii)

In the same way, the real part of $(c + \mathrm{i}s)^6$ gives

$$\cos 6\theta = c^6 - 15c^4s^2 + 15c^2s^4 - s^6,$$

using $\binom{6}{2} = \binom{6}{4} = 15$ and $\mathrm{i}^2 = -1$, $\mathrm{i}^4 = 1$, $\mathrm{i}^6 = -1$. Put $u = c^2$, so $s^2 = 1 - u$:

$$\begin{aligned}\cos 6\theta &= u^3 - 15u^2(1-u) + 15u(1-u)^2 - (1-u)^3 \\ &= u^3 - 15u^2 + 15u^3 + 15u - 30u^2 + 15u^3 - 1 + 3u - 3u^2 + u^3 \\ &= 32u^3 - 48u^2 + 18u - 1 \\ &= 32c^6 - 48c^4 + 18c^2 - 1,\end{aligned}$$

as required.

The equation

Call the left-hand side $p(x) = 16x^6 - 28x^4 + 13x^2 - 1$.

> **Ask yourself** How can the two identities help with a polynomial that isn't either of them?

Compare coefficients. The leading coefficient 16 is half of 32, and the constant -1 is half of -2, which suggests subtracting (i) from (ii):

$$\cos 6\theta - \cos 4\theta = 32c^6 - 56c^4 + 26c^2 - 2 = 2\,p(c).$$

So if $x = \cos\theta$, then

$$p(x) = 0 \iff \cos 6\theta = \cos 4\theta.$$

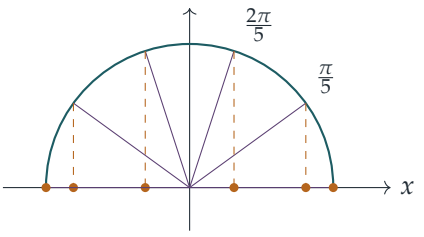
Now $\cos A = \cos B$ exactly when $A = \pm B + 2k\pi$ for some integer k. Here this gives

$$6\theta = 4\theta + 2k\pi \;\Rightarrow\; \theta = k\pi, \qquad \text{or} \qquad 6\theta = -4\theta + 2k\pi \;\Rightarrow\; \theta = \frac{k\pi}{5}.$$

The first family is contained in the second, so the solutions are $\theta = k\pi/5$, and the corresponding values of $x = \cos\theta$ are

$$\cos 0 = 1, \quad \cos\tfrac{\pi}{5}, \quad \cos\tfrac{2\pi}{5}, \quad \cos\tfrac{3\pi}{5} = -\cos\tfrac{2\pi}{5}, \quad \cos\tfrac{4\pi}{5} = -\cos\tfrac{\pi}{5}, \quad \cos\pi = -1.$$

Other values of k repeat these. They are six *distinct* numbers (cosine is strictly decreasing on $[0, \pi]$), and p has degree 6, so these are all the roots of p, real or complex. In particular, the substitution $x = \cos\theta$ only reaches $-1 \le x \le 1$, but we haven't missed any roots outside that interval, since there are none left to find.



These are not yet acceptable answers, since no numerical approximations are allowed and we should give the cosines exactly. Because $x = 1$ is a root, so is $x = -1$ (p is even), and $x^2 - 1$ is a factor. Dividing out (or comparing coefficients),

$$p(x) = (x^2 - 1)(16x^4 - 12x^2 + 1).$$

The quartic factor is a quadratic in x^2:

$$x^2 = \frac{12 \pm \sqrt{144 - 64}}{32} = \frac{12 \pm 4\sqrt{5}}{32} = \frac{3 \pm \sqrt{5}}{8}.$$

Notice that

$$\left(\frac{1+\sqrt{5}}{4}\right)^2 = \frac{6+2\sqrt{5}}{16} = \frac{3+\sqrt{5}}{8}, \qquad \left(\frac{\sqrt{5}-1}{4}\right)^2 = \frac{6-2\sqrt{5}}{16} = \frac{3-\sqrt{5}}{8}.$$

Both values of x^2 are positive, so each gives two real values of x. The real roots of the equation are therefore the six numbers

$$x = \pm 1, \qquad x = \pm\frac{1+\sqrt{5}}{4}, \qquad x = \pm\frac{\sqrt{5}-1}{4}.$$

As a bonus, matching sizes with the list above ($\cos\frac{\pi}{5} > \cos\frac{2\pi}{5} > 0$) gives the well-known values $\cos\frac{\pi}{5} = \frac{1+\sqrt{5}}{4}$ and $\cos\frac{2\pi}{5} = \frac{\sqrt{5}-1}{4}$.

Checks

- At $\theta = 0$ both identities give $1 = 1$; at $\theta = \frac{\pi}{2}$, (i) gives $1 = 1$ and (ii) gives $-1 = -1$.
- The coefficients of p add to $16 - 28 + 13 - 1 = 0$, confirming the root $x = 1$.
- Numerically, $\frac{1+\sqrt{5}}{4} \approx 0.80902 = \cos 36^\circ$ and $\frac{\sqrt{5}-1}{4} \approx 0.30902 = \cos 72^\circ$, and p vanishes at both (to rounding error).

Common slips

- Forgetting the solutions from $6\theta = -4\theta + 2k\pi$, and finding only $x = \pm 1$.
- Leaving answers as $\cos\frac{\pi}{5}$ and so on, or as decimals: the question wants exact surd forms.
- Not explaining why there are no roots with $|x| > 1$: the degree count settles it.

Beyond STEP

The polynomial expressing $\cos n\theta$ in terms of $\cos\theta$ is the Chebyshev polynomial T_n, so (i) and (ii) say $T_4(x) = 8x^4 - 8x^2 + 1$ and $T_6(x) = 32x^6 - 48x^4 + 18x^2 - 1$. Chebyshev polynomials are central in numerical analysis, where their roots give the best points for

polynomial interpolation. The appearance of $\sqrt{5}$ in $\cos\frac{\pi}{5}$ reflects the geometry of the regular pentagon and the golden ratio. Needham's *Visual Complex Analysis* is a lovely place to see de Moivre's theorem and roots of unity geometrically.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.